DTC P2227:00 [SKYACTIV-D 1.5]

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DTC P2227:00	BARO sensor circuit range/performance problem
DETECTION CONDITION	• With the following conditions met, a condition has continued for **1.8 s** within **7 to 9 s** after the ignition was switched OFF in which difference between the barometric pressure and intake air pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more** and difference between the barometric pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — The following DTCs are not detected: • MAP sensor: P0106:00 • Intake shutter valve: P1589:00 • BARO sensor: P2228:00, P2229:00 **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Limits the engine torque or the upper limit of the engine speed. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • Malfunction in the following sensors — MAP sensor • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• BARO sensor (built-into PCM) malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Perform the Freeze Frame PID Data Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the DTC P2227:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
2	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
3	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
4	**VERIFY OTHER RELATED DTCs** • Display the DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Has any DTC related to the following sensors been stored? — MAP sensor	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT BARO SENSOR** • Inspect the BARO sensor. (See BAROMETRIC PRESSURE (BARO) SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine and idle it. • Switch the ignition to off and wait for **10 s or more**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	• Inspect the following sensors and perform Step 6 again. — MAP sensor (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR INSPECTION [SKYACTIV-D 1.5].) • If the malfunction recurs, replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.